

17-2e Why People Sometimes Cooperate

The prisoners' dilemma shows that cooperation is difficult. But is it impossible? Not all prisoners, when questioned by the police, decide to turn in their partners in crime. Cartels sometimes manage to maintain collusive arrangements, despite the incentive for individual members to defect. Very often, players can solve the prisoners' dilemma because they play the game not once but many times.

To see why cooperation is easier to enforce in repeated games, let's return to our duopolists, Jack and Jill, whose choices were given in [Figure 2](#). Jack and Jill would like to agree to maintain the monopoly outcome in which each produces 30 gallons. Yet, if Jack and Jill are to play this game only once, neither has any incentive to live up to this agreement. Self-interest drives each of them to renege and choose the dominant strategy of 40 gallons.

Now suppose that Jack and Jill know that they will play the same game every week. When they make their initial agreement to keep production low, they can also specify what happens if one party reneges. They might agree, for instance, that once one of them reneges and produces 40 gallons, both of them will produce 40 gallons forever after. This penalty is easy to enforce because if one party produces at a high level, the other has every reason to do the same.

The threat of this penalty may be all that is needed to maintain cooperation. Each person knows that defecting would raise his or her profit from \$1,800 to \$2,000. But this benefit would last for only one week. Thereafter, profit would fall to \$1,600 and stay there. As long as the players care enough about future profits, they will choose to forgo the one-time gain from defection. Thus, in a game of repeated prisoners' dilemma, the two players may well be able to reach the cooperative outcome.

Case Study: The Prisoners' Dilemma Tournament

Imagine that you are playing a game of prisoners' dilemma with a person being "questioned" in a separate room. Moreover, imagine that you are going to play the game with this other person not once but many times. Your score at the end of the game is the total number of years in jail. You would like to minimize this score. What strategy would you play? Would you begin by confessing or remaining silent? How would the other player's actions affect your subsequent decisions about confessing?

Repeated prisoners' dilemma is a complicated game. To encourage cooperation, players must penalize each other for not cooperating. Yet the strategy described earlier for Jack and Jill's water cartel—defect forever as soon as the other player defects—is not very forgiving.

In a game repeated many times, a strategy that allows players to return to the cooperative outcome after a period of noncooperation may be preferable.

To see what strategies work best, political scientist Robert Axelrod held a tournament. People entered by submitting computer programs designed to play repeated prisoners' dilemma. Each program then played the game against all the other programs. The "winner" was the program that received the fewest total years in jail.

The winning program turned out to be a simple strategy called *tit-for-tat*. According to tit-for-tat, a player should start by cooperating and then do whatever the other player did last time. Thus, a tit-for-tat player cooperates until the other player defects; then she defects until the other player cooperates again. In other words, this strategy starts out friendly, penalizes unfriendly players, and forgives them if warranted. To Axelrod's surprise, this simple strategy did better than all the more complicated strategies that people had sent in.

The tit-for-tat strategy has a long history. It is essentially the classic strategy of "an eye for an eye, a tooth for a tooth." The prisoners' dilemma tournament suggests that this may be a good rule of thumb for playing some of the games of life.

QuickQuiz

5. The prisoners' dilemma is a two-person game illustrating that
 - a. the cooperative outcome could be worse for both people than the Nash equilibrium.
 - b. even if the cooperative outcome is better than the Nash equilibrium for one person, it might be worse for the other.
 - c. even if cooperation is better than the Nash equilibrium, each person might have an incentive not to cooperate.
 - d. rational, self-interested individuals will naturally avoid the Nash equilibrium because it is worse for both of them.
6. Two people facing the prisoners' dilemma may cooperate if
 - a. they recognize that the Nash equilibrium is worse for both people than the cooperative equilibrium.
 - b. they will play the game repeatedly and expect noncooperation to be met with future retaliation.
 - c. each chooses the strategy that is best for herself, given what the other person is doing.

d. each realizes that the strategy she chooses is not known to the other until the outcome is realized.

Chapter 17: Oligopoly: 17-2e Why People Sometimes Cooperate

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